

XINYI (CINDY) ZHOU

☎ 213-272-3352 ✉ xinyi.zhou223@gmail.com

🌐 www.linkedin.com/in/xinyi-zhou-cindy/ 🏠 github.com/CindyChow123 🌐 cindychow123.github.io/

EDUCATION

University of Southern California

M.S. in Computer Science; GPA: 4.0/4.0

January 2023 – December 2024

Los Angeles, CA

Southern University of Science and Technology

B.E. in Computer Science and Technology; **Outstanding Graduate Award**; GPA: 3.75/4.0

September 2018 – July 2022

Shenzhen, China

TECHNICAL SKILLS

Languages: TypeScript, Python, SQL, JavaScript, Java, C#, C++, HTML/CSS

Full-Stack Architecture: Next.js (SSR), Node.js, React, Vue.js; FastAPI, REST APIs, XMPP Integration, Feature Flags (Unleash), Redis

Data/Infra: BigQuery, Snowflake, PostgreSQL, Amplitude, ETL; AWS (ECS, Lambda, S3, RDS/Aurora, CloudFront), Docker, CI/CD

LLMs: PyTorch, Hugging Face Transformers, DeepSpeed, Prompt Engineering, LLM Evaluation, MCP

Developer Operations & Observability: Better Stack, Sentry, Jira, Jenkins, Kibana, Slack Webhooks/Bots, Claude Code, Git Worktrees, AI Agentic Workflows for Automated Error Detection & Debugging

WORK EXPERIENCE

Software Engineer (Backend / AI Product) | Channel AI | Menlo Park, US

Feb 2025 – Present

- Spearheaded an **LLM-powered** personalization feature in **TypeScript**; extended the **PostgreSQL** schema to dynamically inject targeted user context into model prompts, delivering a **+45% revenue uplift** and a 4% increase in daily messaging for mature cohorts within 4 days
- Implemented a highly available **LLM serving pipeline** by deploying **HuggingFace** models to GPU servers and configuring a **LiteLLM** proxy gateway, enabling secure backend access with automated retry and fallback logic while processing up to **27,000 requests** per hour.
- Engineered an automated Trust & Safety pipeline integrating **external ML classifiers**; collaborated with Product and Operations teams to implement a **rules-based evaluation engine** that reduced manual review by **78%** by automatically handling definitive cases, routing failed evaluations to human queues, and logging high-risk content for future policy tuning
- Architected a fault-tolerant, hourly **TypeScript ETL pipeline** routing Amplitude data to **BigQuery**; implemented windowed backfills and automated retries to handle late-arriving events and maintain recommendation engine freshness
- Created an autonomous Slack bot using **Claude** to surface daily **Sentry/Better Stack** errors and assign owners; enabled conversational debugging that reduced manual log investigation by **~80% per issue** and automatically generated PRs for rapid remediation
- Orchestrated a **time zone-aware**, asynchronous payout pipeline in **TypeScript** to distribute prior-day rewards for **4,000+** creators; implemented **idempotent** retry mechanisms and database state tracking to ensure 100% financial accuracy and system reliability
- Developed scalable backend microservices on **AWS (ECS)** for mobile and web clients, implementing **XMPP** for real-time messaging and **Redis** for session management; optimized Amazon **Aurora PostgreSQL** queries and indexing to significantly reduce latency and enhance data reliability
- Designed and executed **zero-downtime schema migrations** via **Prisma**; utilized **parallel-writing** and **batched backfill scripts** to evolve the **Aurora PostgreSQL** schema without service degradation

Software Engineer Intern | Veeva Systems, Vault Platform Reporting | Pleasanton, United States

May 2024 – August 2024

- Maintained **Spring Boot microservices** and **React** UIs within a high-scale cloud reporting platform; resolved **11 defects** and expanded automated unit and integration test coverage using **JUnit** and **Jest** via **Jenkins** CI/CD pipelines
- Implemented a real-time data freshness indicator by integrating **React** front-end with **downstream microservice APIs**, ensuring accurate visibility into asynchronous data processing states
- Resolved race conditions and key collisions in a **distributed multi-tier caching architecture**, ensuring data consistency across service instances for **1,000+** customers

Machine Learning (NLP) Engineer Intern | UBTECH Robotics | Shenzhen, China

May 2023 – August 2023

- Fine-tuned LLM Vicuna chatbots (7B, 13B, and 33B) with **DeepSpeed ZeRO** and **LoRA** on a V100 GPU cluster for more engaging human-robot interactions in reception and tour guiding, obtaining an increase of 29.74% in user preference and 22 in Elo rating
- Led design of **Python** workflow for conversation data generation, cleaning, and formatting based on **prompt engineering** and ChatGPT API; system applied to at least 3 NLP projects, resulting in a 20x efficiency improvement in fine-tuning data collection
- Deployed a Chatbot implemented by **Gradio**, **Uvicorn**, and **FastAPI** on a cloud service for users to evaluate fine-tuned models

Research Assistant of Dr.Luo Tao | SUSTech | Shenzhen, China

August 2022 – December 2022

- Collaborated with 8 UX designers to design and create innovative MR/VR user interaction demos with **Unity** on MR/VR devices.
- Developed **C#** and **Python** scripts to automatically collect and analyze user hand motion data for user experience testing.
- Analyzed semi-structured interview materials about the industry's application of the IxDL language, extracting actionable insights for promotion within technical teams.

PUBLICATIONS

Xinyi Zhou, Zeinabsadat Saghi, Sadra Sabouri, Rahul Pandita, Mollie McGuire, Souti Chattopadhyay. “Cognitive Biases in LLM-Assisted Software Development.” *ICSE '26*, 2026. arXiv:2601.08045

Sadra Sabouri, Philipp Eibl, **Xinyi Zhou**, Morteza Ziyadi, Nenad Medvidovic, Lars Lindemann, Souti Chattopadhyay. “Trust Dynamics in AI-Assisted Development: Definitions, Factors, and Implications.” *ICSE '25*, 2025. doi:10.1109/ICSE55347.2025.00199

Xinyi Zhou, Louying Hao, Qiushi Nie, Yingquan Zhou, Lihui Wang, Yan Hu, Jiang Liu. “A Novel Multi-focus Fusion Network for Retinal Microsurgery.” *ISBI*, 2022. doi:10.1109/ISBI52829.2022.9761531

PROJECTS

Cognitive Biases in LLM-Assisted Software Development | *Human-AI Interaction Researcher* **January 2024 – October 2025**

- Primary researcher for an ICSE 2026 paper on cognitive biases in LLM-assisted software development; led literature review, study design, recruitment, execution, and synthesis (arXiv:2601.08045)
- Ran mixed-methods research with developers (observations: n=14; surveys: n=22), combining behavioral coding and statistical analysis to measure bias in human-LLM workflows
- Built a validated taxonomy of 15 bias categories (systematic analysis of 90 cognitive biases) and translated findings into concrete product/engineering guidelines for designing and evaluating LLM-driven systems with bias-mitigation guardrails
- Measured bias prevalence in human-LLM workflows (48.8% biased actions; 56.4% tied to developer-LLM interactions), using results to define guardrail requirements and evaluation metrics

Color Splash | *Game Developer* **January 2024 – May 2024**

- Designed and developed a **2D Unity/C# platformer** in a team of 6 where players switch colors to unlock abilities, solve environmental puzzles, avoid color-specific hazards, and break barriers through projectile mechanics
- Implemented Firebase gameplay analytics to track player attempts, color usage, skill frequency, and death events, using Python visualizations to identify level design issues and validate findings with playtest feedback

eBay Search Platform | *Full-Stack Developer* **September 2023 – December 2023**

- Crafted responsive web UIs with **Angular**, **Bootstrap**, **HTML/CSS**, and **JavaScript**, and mobile UIs on Android using **Kotlin**
- Enhanced backend functionality with **NodeJS**, **Express**, **MongoDB**, deployed on **AWS**

Color Correction WeChat Mini Program | *Front-end Developer; Project Manager* **February 2021 – May 2021**

- Implemented the front-end using **HTML/CSS** and developed a **JavaScript** algorithm for color space conversion that simulated the visual experience of individuals with color blindness to promote equality for them.
- Collaborated with **Tencent** Charity Foundation to write product requirement documents and managed team member's development schedule to ensure the product was finished on time